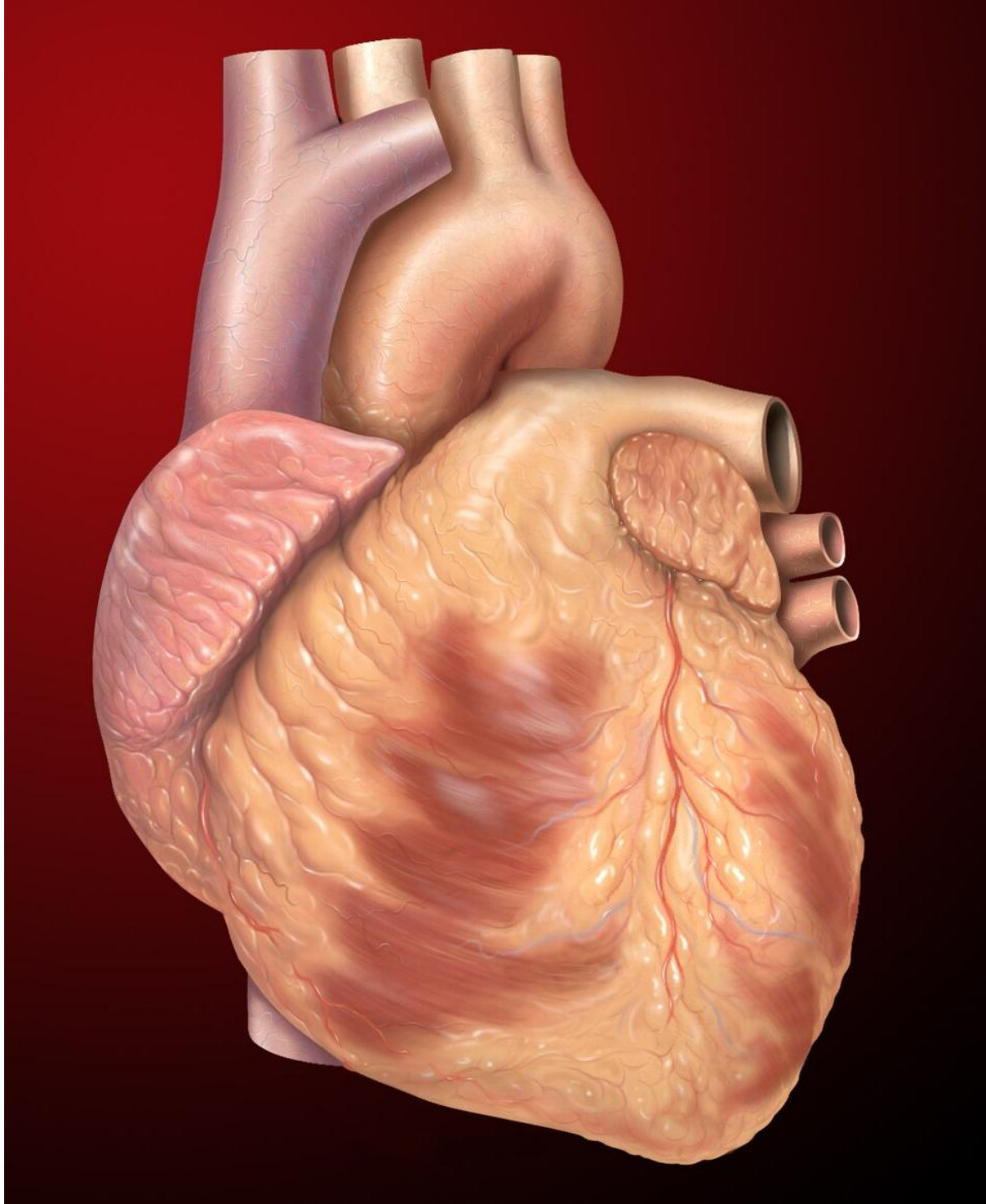
DP Biology – B3.2 (HL extension)

Tissue fluid, lymph and the heart

- Which adaptations help animals and plants move fluids around?
- How does transport in animals compare with transport in plants?

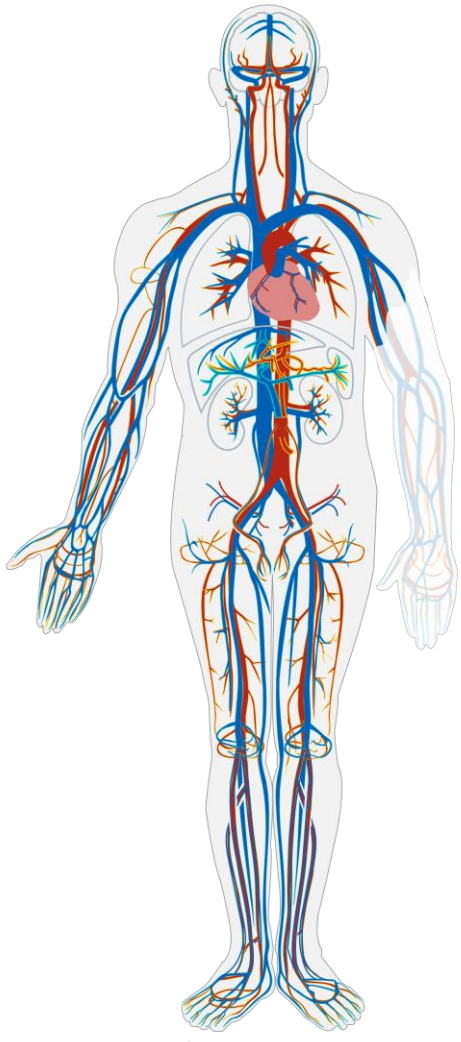
HL statements B3.2.11–16. Textbook pp. 310–321.
Image: the human heart. BruceBlaus, CC BY 3.0.



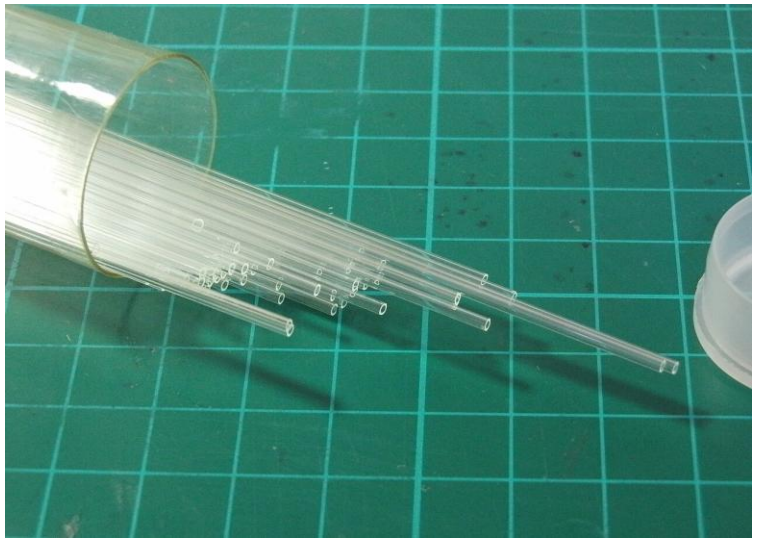
Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.2.11–13	Explain how tissue fluid forms, exchanges with cells and drains into lymph
2	B3.2.14–15	Compare single and double circulation, and explain how the heart is adapted to pump under pressure
3	B3.2.16	Describe the cardiac cycle, and interpret pressure graphs and blood pressure data

From the heart to every cell



Circulatory system. Public domain.



A capillary with red blood cells. CC BY-SA 3.0.

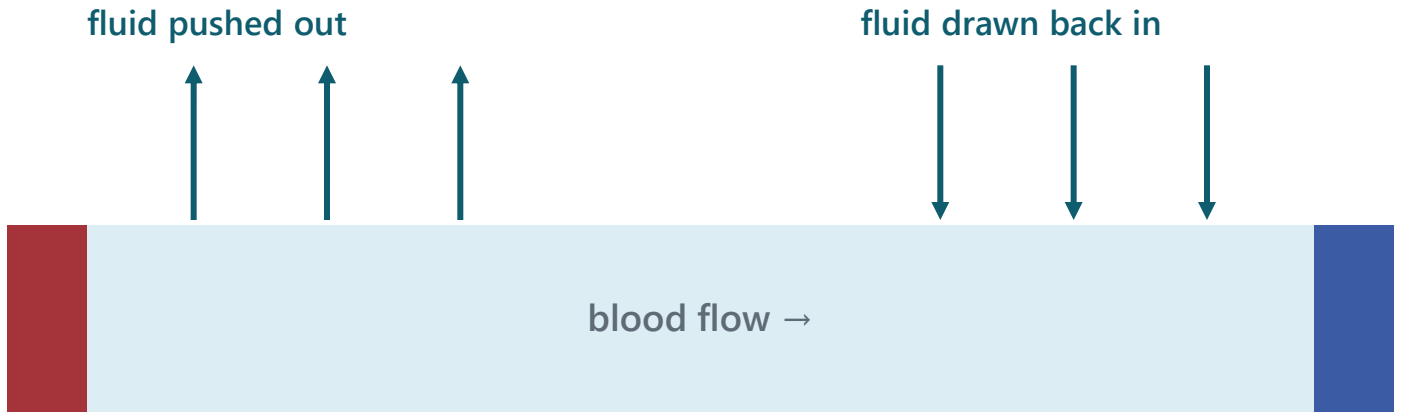
This chapter

Class 1: tissue fluid and lymph. Class 2: circulation and the heart. Class 3: one heartbeat.

The heart pumps blood out through arteries to **capillaries**, so narrow that red blood cells pass in single file.

But blood never touches your cells directly. Between them is a thin layer of liquid: **tissue fluid**. Everything a cell gets or gets rid of passes through it.

How tissue fluid forms



body cells bathed in tissue fluid

At the **arteriole end**, high blood pressure forces plasma out through gaps in the capillary wall: **pressure filtration**.

Proteins and cells are too big and stay in.

At the **venule end**, pressure is low, so most of the fluid moves back in.

Two forces decide the direction

B3.2.11

	Arteriole end	Venule end
Blood pressure: pushes out	high (about 4.5 kPa)	low (about 2 kPa)
Pull of plasma proteins: draws in	about 3.5 kPa	about 3.5 kPa
Net movement	out	in

Why the pull stays the same

Plasma proteins are too big to leave the capillary, so their concentration, and the water they draw back by osmosis, hardly changes. Only the blood pressure falls along the capillary.

Plasma and tissue fluid compared

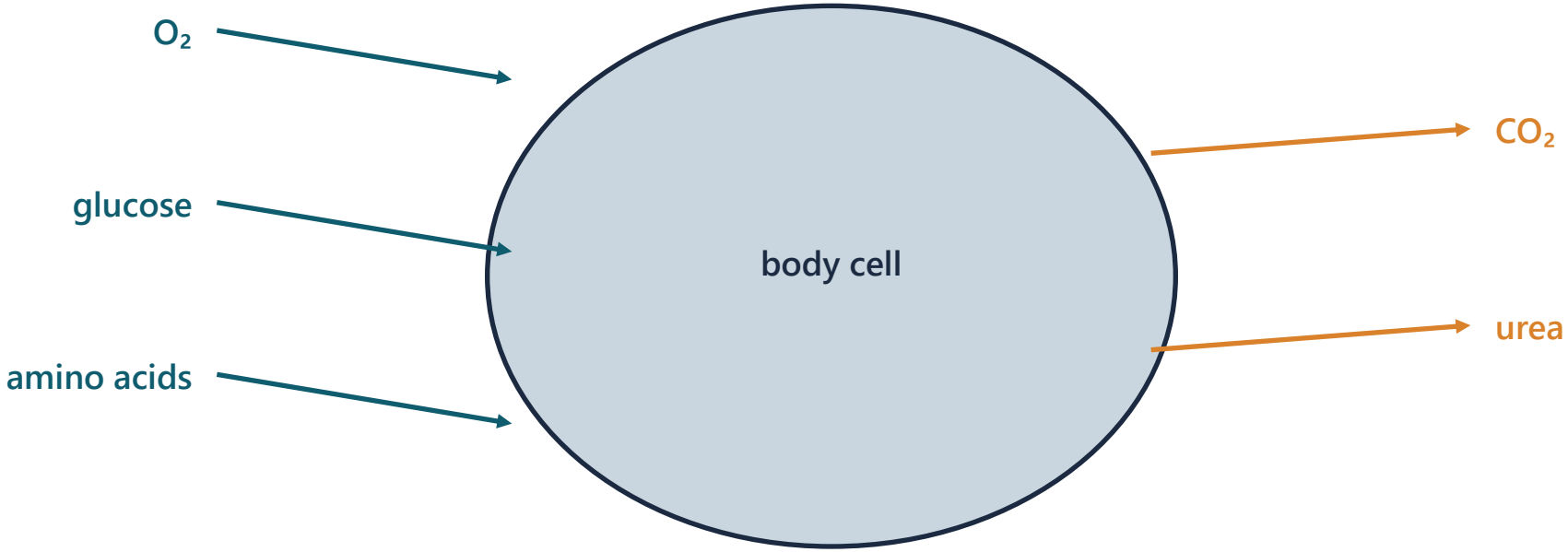
B3.2.12

	Plasma	Tissue fluid
Water, ions, glucose, amino acids	yes	yes
Large proteins	lots	very little
Red blood cells	yes	no
Oxygen	more	less: used by cells
Carbon dioxide, urea	less	more: made by cells

Tissue fluid is plasma without its large proteins and red blood cells.

Exchange between tissue fluid and cells

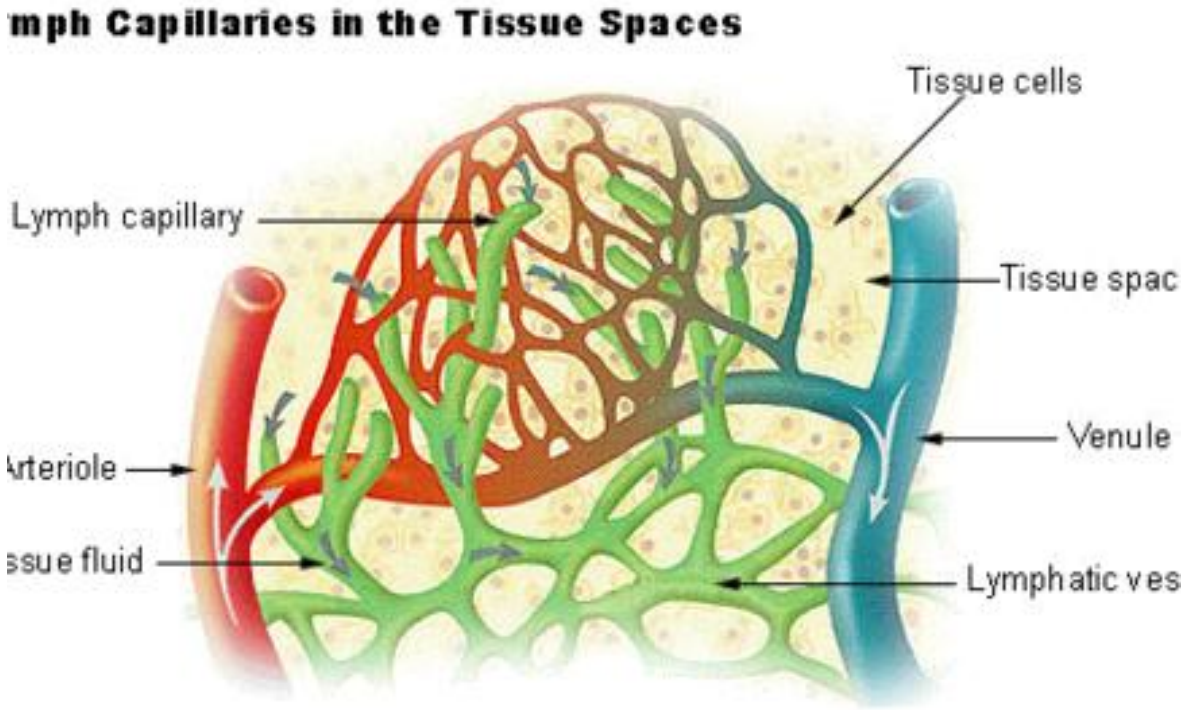
B3.2.12



Substances cross the cell membrane by diffusion, facilitated diffusion or active transport. The distance from capillary to cell is tiny, so exchange is fast.

The fluid that returns to the capillary carries the wastes away.

Draining extra fluid into lymph



Lymph capillaries among tissue cells. SEER, NCI, public domain.

More fluid leaves the capillaries than returns. The extra enters **lymph capillaries**, which have gaps between their cells and are closed at one end.

The fluid is now **lymph**. **Valves** keep it moving one way, back to veins near the heart.

If drainage fails

Fluid builds up and the tissue swells: oedema. Lymph also passes through lymph nodes, where white blood cells filter it.

Practice: why do tissues swell?

B3.2.11–13

High blood pressure	Starvation	Blocked lymph vessel
Blood pressure in capillaries is raised.	Little protein is eaten, so plasma protein falls.	A parasite blocks a lymph duct in the leg.

1. For each case, explain why tissue fluid builds up.
2. Which case affects the arteriole end, which the venule end, and which the drainage?

Check-in: class 1

B3.2.11–13

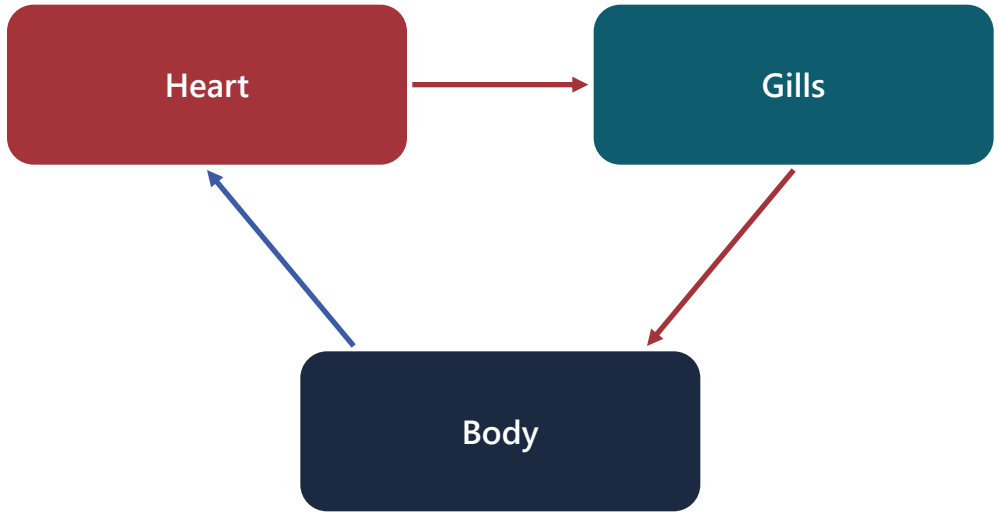
1. Explain why fluid leaves the capillary at the arteriole end.
2. Explain why most of it returns at the venule end.
3. State two differences between plasma and tissue fluid.
4. Describe how substances pass between tissue fluid and cells.
5. Outline what happens to tissue fluid that does not return to the capillary.

How to use this

Answer without your notes, then check against slides 4–9.

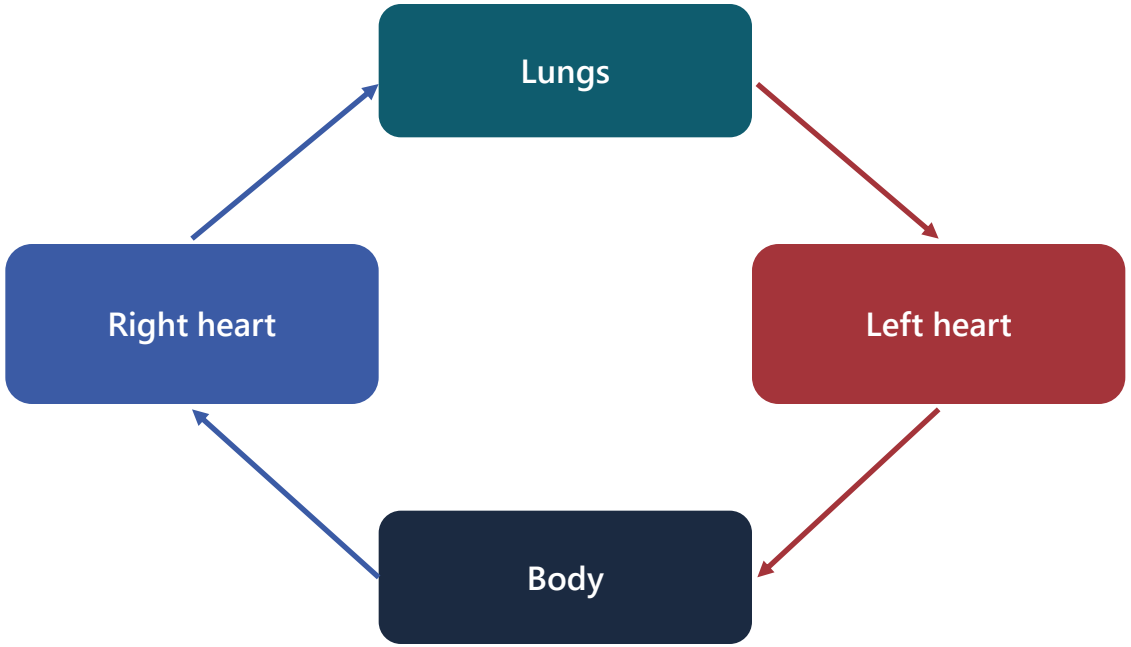
Single and double circulation

Bony fish: single circulation



Through the heart once per circuit

Mammals: double circulation



Through the heart twice per circuit

Draw it

Simple circuit diagrams like these are all you need: the order of organs and the direction of flow.

Why two circuits?

B3.2.14

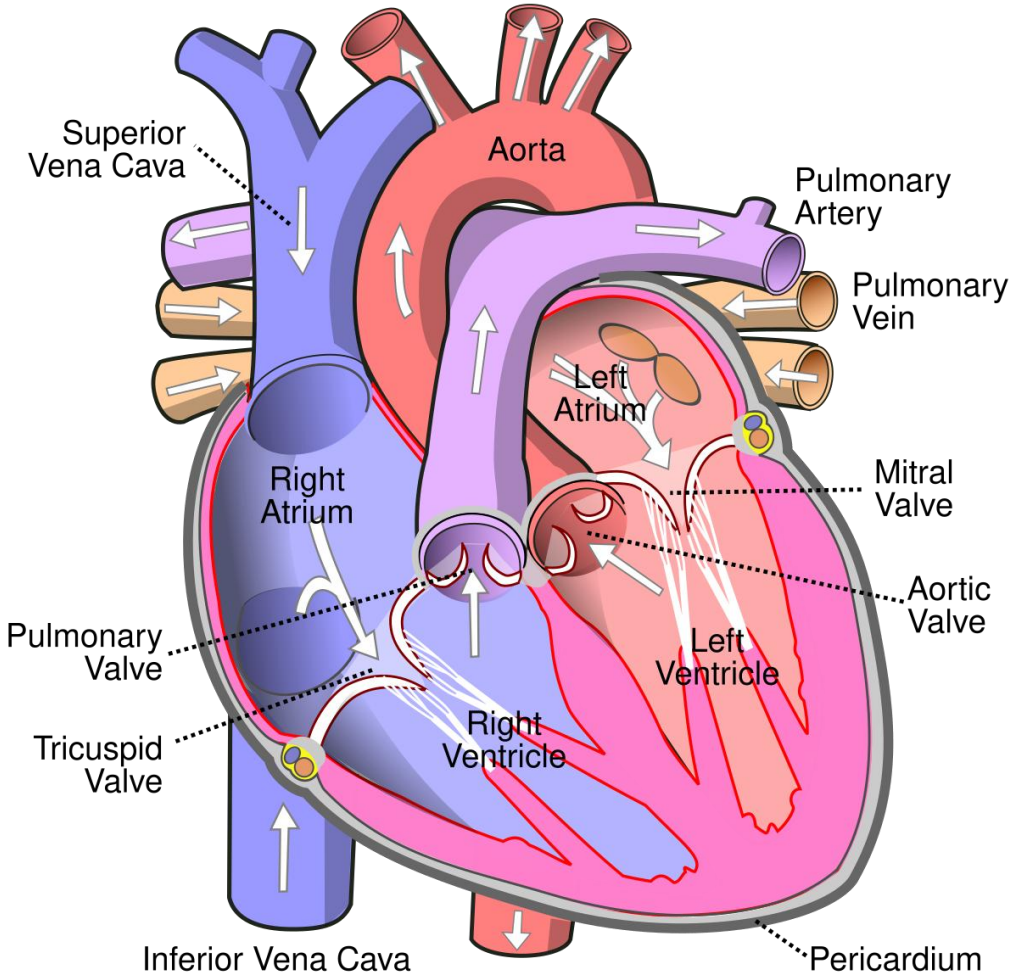
	Bony fish	Mammal
Chambers	1 atrium, 1 ventricle	2 atria, 2 ventricles
After the gills or lungs	straight on to the body	back to the heart to be re-pumped
Pressure to the body	low	high

Why it matters

Narrow gill or lung capillaries cut blood pressure. Re-pumping lets mammals send blood to the body fast and at high pressure, supporting a high metabolic rate and a constant body temperature.

The heart in frontal section

B3.2.15



vena cava

↓

right atrium

↓

right ventricle

↓

pulmonary artery: lungs

↓

pulmonary vein

↓

left atrium

↓

left ventricle

↓

aorta: body

Valves between each atrium and ventricle, and at the base of each artery, keep blood moving one way.

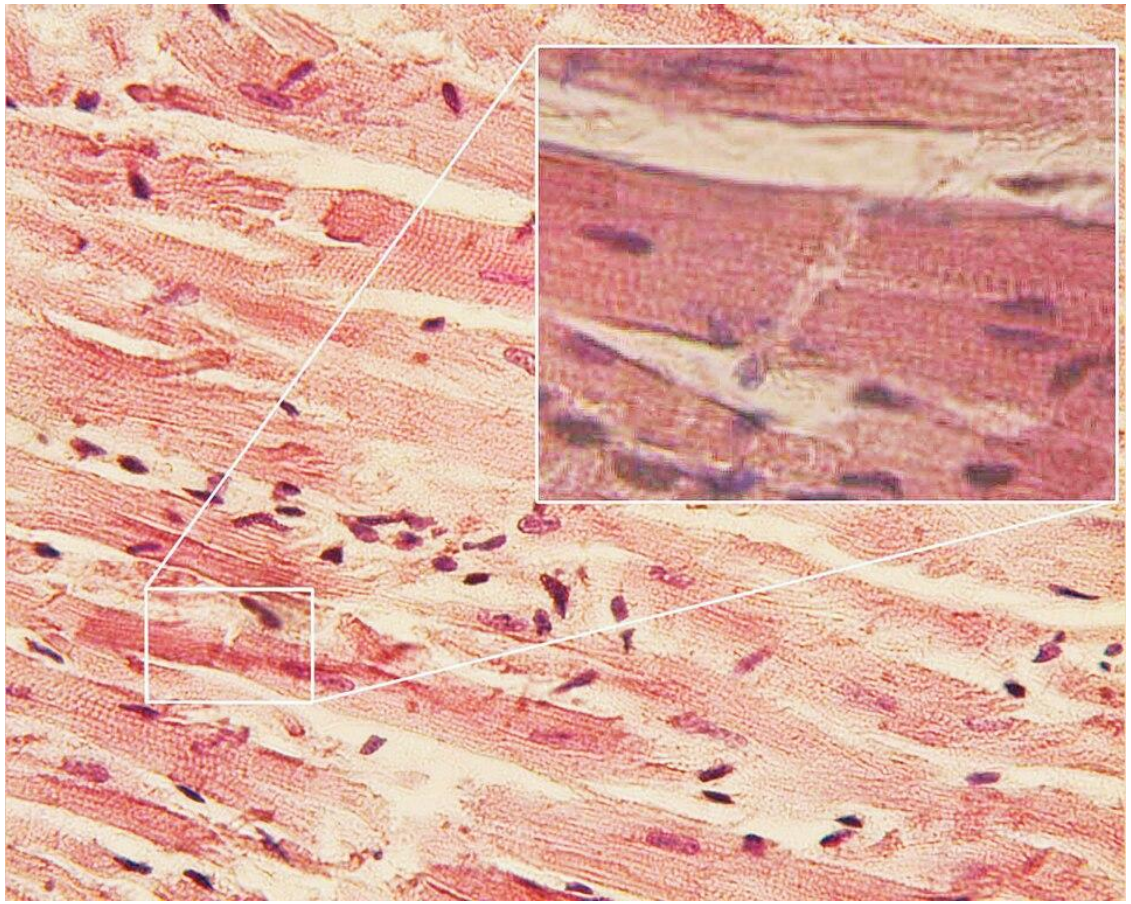
The heart in frontal section. Wapcaplet, CC BY-SA 3.0.

How the heart delivers blood under pressure

B3.2.15

Structure	How it helps
Thick ventricle walls	strong contraction; the left is thickest (whole body)
Thin atrium walls	push blood only into the ventricles
AV valves	stop backflow into the atria when ventricles contract
Semilunar valves	stop backflow from the arteries
Septum	keeps oxygenated and deoxygenated blood apart
Coronary arteries	supply the heart muscle itself

Cardiac muscle



Cardiac muscle; inset: an intercalated disc. Girod and Becker, CC BY 2.5.

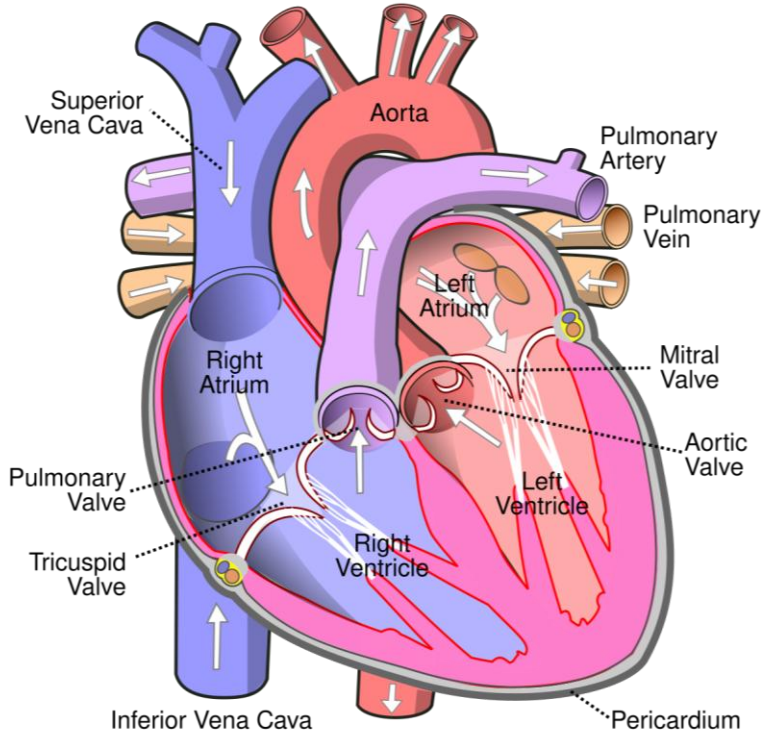
Cells are **branched** and joined by **intercalated discs**, so impulses spread fast and the cells contract together.

Many mitochondria: it never rests.

Myogenic: each beat starts in the muscle itself, not from a nerve.

Practice: trace the blood

B3.2.14–15



Heart in section. CC BY-SA 3.0.

1. Trace a red blood cell from the left big toe back to the left big toe. Name each chamber, valve and vessel.
2. State how many times it passes through the heart.
3. Explain why the left ventricle wall is thicker than the right.
4. Suggest why a hole in the septum leaves a baby short of oxygen.

Check-in: class 2

B3.2.14–15

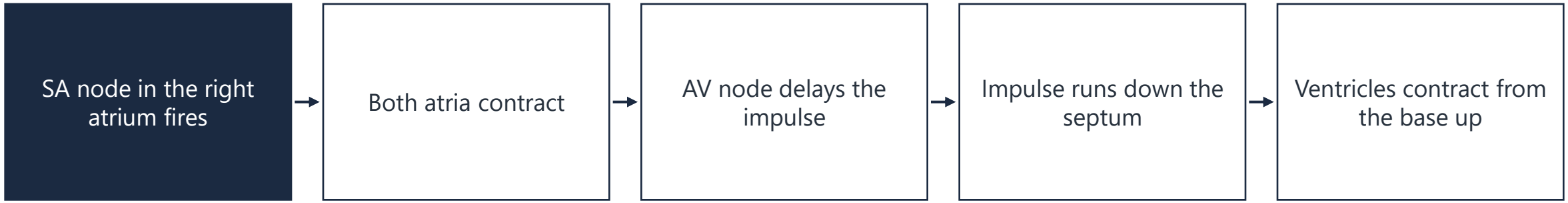
1. Distinguish between single and double circulation.
2. Explain one advantage of double circulation.
3. State the role of the AV and semilunar valves.
4. Explain why cardiac muscle cells are branched.
5. Define myogenic.

How to use this

Answer without your notes, then check against slides 11–16.

Starting each heartbeat

B3.2.16



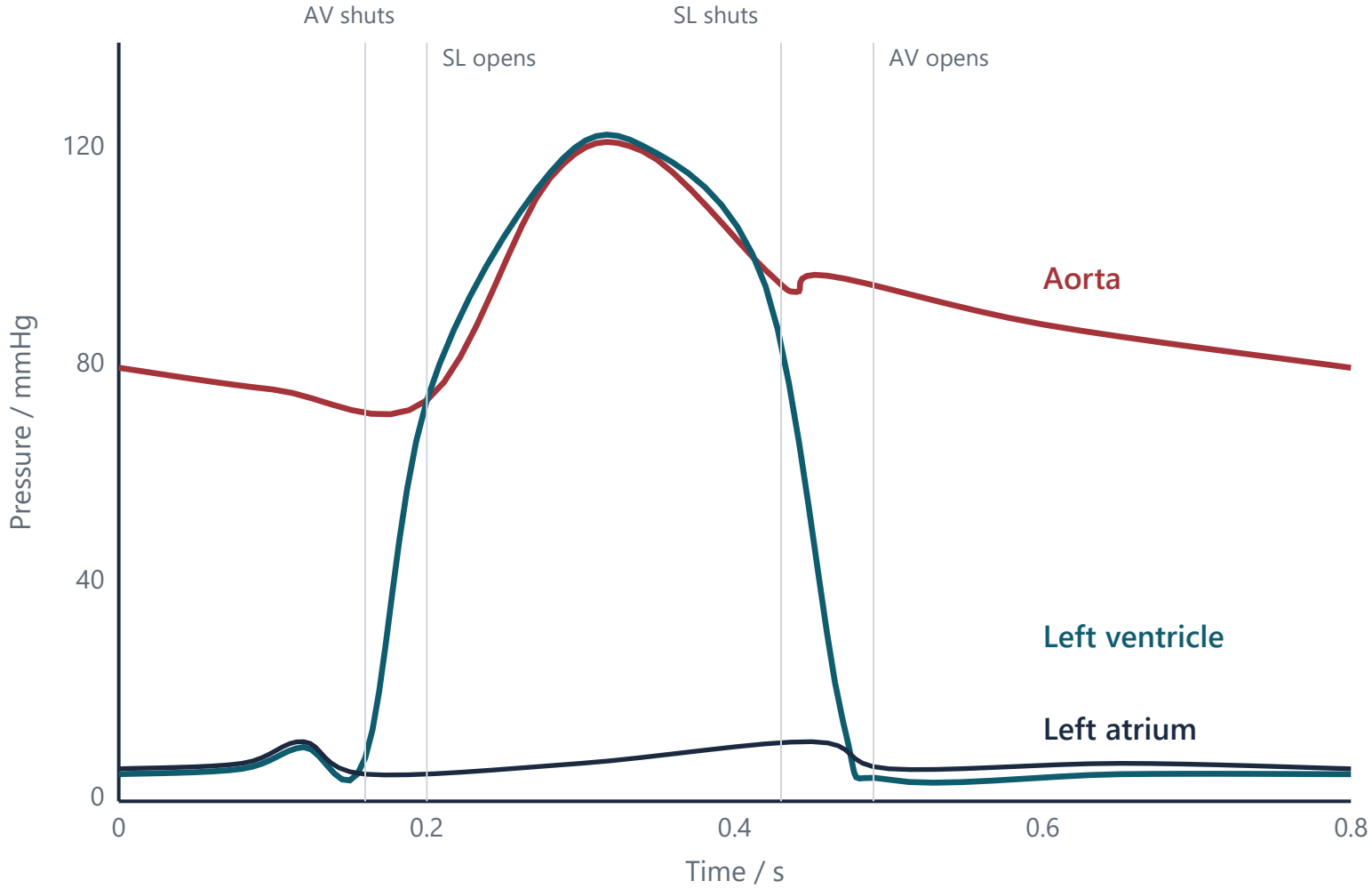
Why the delay?

It gives the atria time to finish emptying into the ventricles before these contract.

Rate

The SA node (the pacemaker) sets about 60–100 beats per minute. Nerves and adrenaline speed it up or slow it down.

Pressures in the left side of the heart



Schematic, one beat at 75 beats per minute. AV: atrioventricular valve. SL: semilunar valve.

Valves open and close because of **pressure differences**: a valve opens when the pressure behind it is higher than in front.

Systole: contraction. **Diastole**: relaxation.

Stages of the cardiac cycle

B3.2.16

Stage	What happens	Valves
Atrial systole	atria contract, topping up the ventricles	AV open, SL shut
Ventricular systole begins	ventricle pressure rises; AV valves slam shut (first sound)	both shut
Ventricular systole	pressure passes artery pressure; blood ejected	AV shut, SL open
Diastole begins	ventricles relax; SL valves shut (second sound)	both shut
Diastole	chambers fill from the veins	AV open, SL shut

Rule: blood always flows from higher to lower pressure.

Systolic and diastolic blood pressure

B3.2.16



Blood pressure cuff. Public domain.

Systolic: the highest pressure in the arteries, as the ventricles eject blood.
Diastolic: the lowest, while the heart relaxes.

Written systolic/diastolic, for example **120/80 mmHg**. Above about 140/90 at rest is high blood pressure.

Think

Why does pressure in the aorta never fall to zero between beats?

Working with data: blood pressure and exercise

B3.2.16

Blood pressure of one student at rest and during exercise on a bike. These values are invented for practice; the pattern is realistic.

Activity	Systolic / mmHg	Diastolic / mmHg
Rest	118	76
Light cycling	140	78
Hard cycling	175	80
5 min after	125	76

1. Describe how systolic and diastolic pressure change.
2. Explain the rise in systolic pressure.
3. Calculate the pulse pressure at rest and in hard cycling.

Common mistakes to avoid

Not this	But this
Tissue fluid contains red blood cells.	It has no red cells and little protein.
All tissue fluid returns to capillaries.	The excess drains into lymph.
The left side pumps oxygenated and the right deoxygenated, one after the other.	Both sides contract at the same time.
Valves are opened by muscles.	Valves open and close because of pressure differences.

Check that you can...

1. **Explain** how tissue fluid forms and returns to capillaries
2. **Compare** plasma and tissue fluid
3. **Outline** how lymph capillaries drain excess fluid
4. **Compare** single and double circulation, with circuit diagrams
5. **Explain** how the heart is adapted to pump blood under pressure
6. **Describe** the cardiac cycle and **interpret** pressure graphs
7. **Distinguish** systolic and diastolic blood pressure

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why fluid leaves a capillary at the arteriole end and returns at the venule end.
2. State two differences between the circulation of a bony fish and of a mammal.
3. Explain why the semilunar valves close at the start of diastole.

Image credits

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